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CSE260 Quiz 4

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Section 3

Q1. In a high-tech smart washing machine, an LED indicator is used to show the status of the washing cycle. The LED follows this sequence to represent the different stages of operation:

- Blue: Washing started.
- Green: Washing in progress.
- Yellow: Rinsing in progress.
- Red: Machine idle or needs user attention.

The cycle repeats in the following order: Blue → Green → Yellow → Green → Red → Blue.
Design and draw the circuit diagram for the above scenario using SR Flip-Flops.

Blue → Green → Yellow → Green → Red → Blue

00

01

10

01

11

00

Blue - 00

Green - 01

Yellow - 10

Red - 11

Let,

Green ⁰¹ → Yellow ¹⁰

↓
Red

state table:

x	A	B	A ⁺	B ⁺	S _A	R _A	S _B	R _B
0	0	0	0	1	0	x	1	0
0	0	1	1	0	1	0	0	1
0	1	0	0	1	0	1	1	0
0	1	1	0	1	0	1	0	1
1	0	0	0	1	0	x	1	0
1	0	1	1	0	1	0	x	0
1	1	0	1	1	0	1	1	0
1	1	1	0	0	0	1	0	1

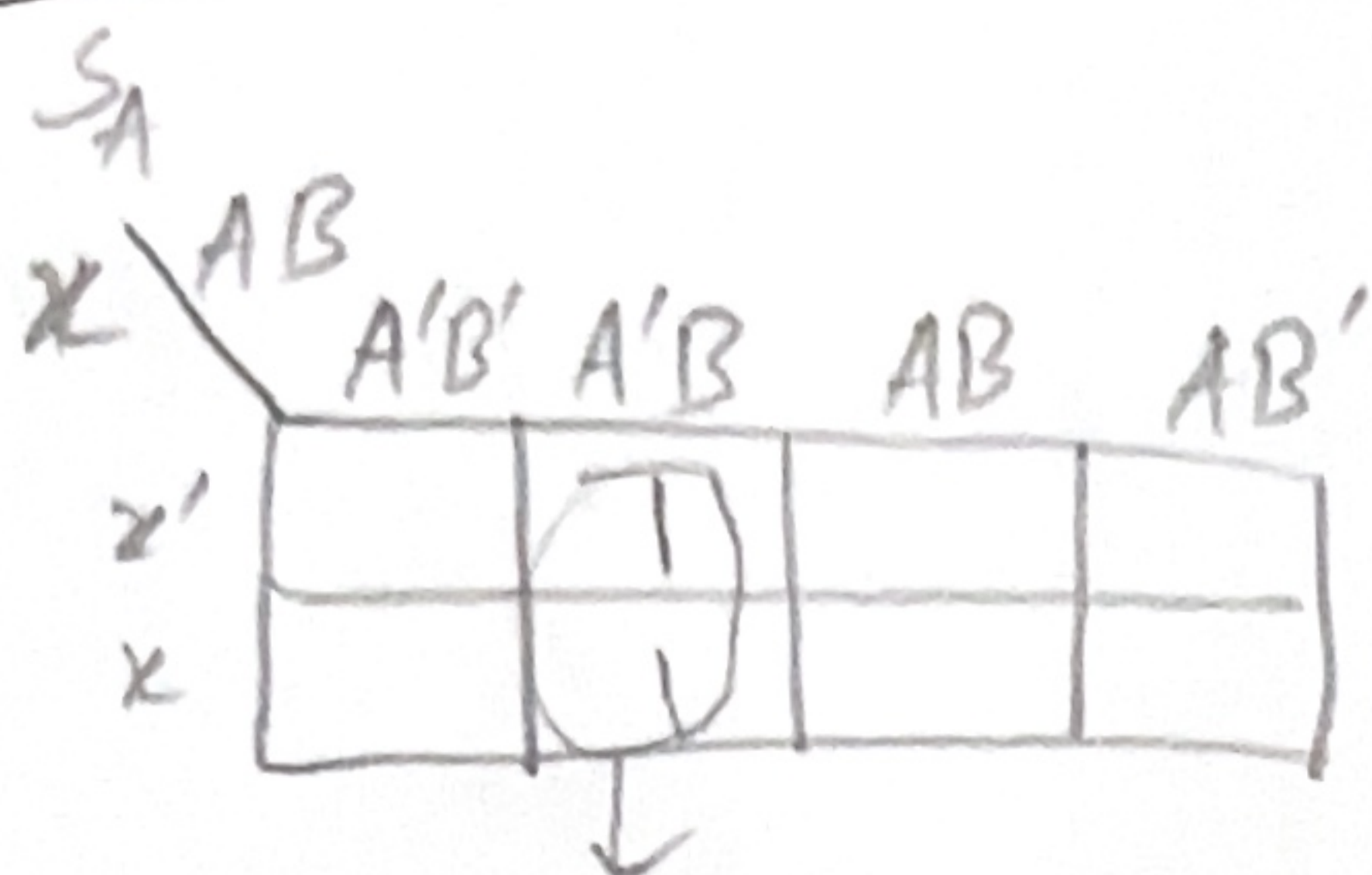
Charac. table (SR)

Q	S	R	Q ⁺
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	not used
1	0	0	1
1	0	1	0
1	1	0	not used
1	1	1	not used

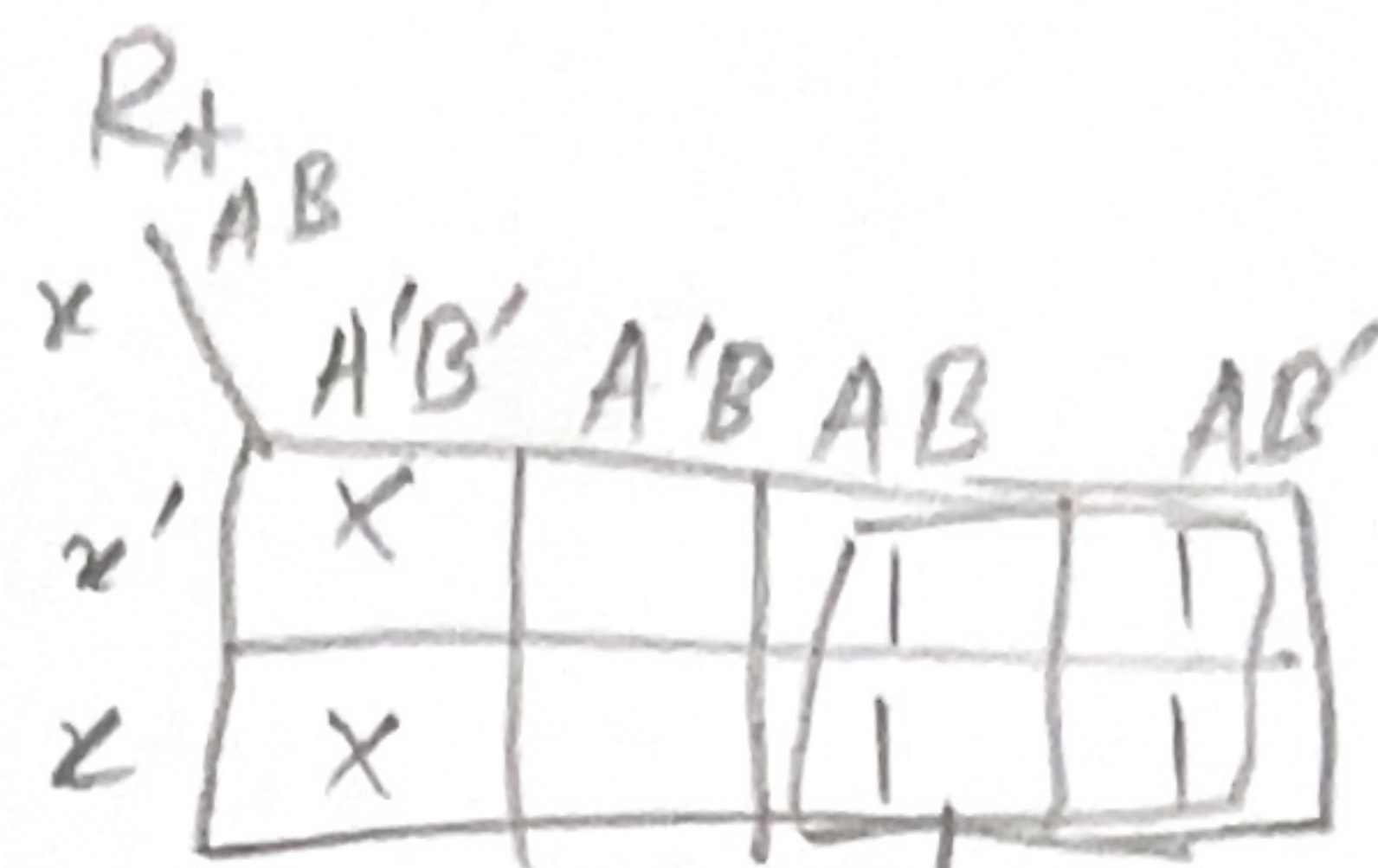
Excitation (SR)

Q	Q ⁺	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

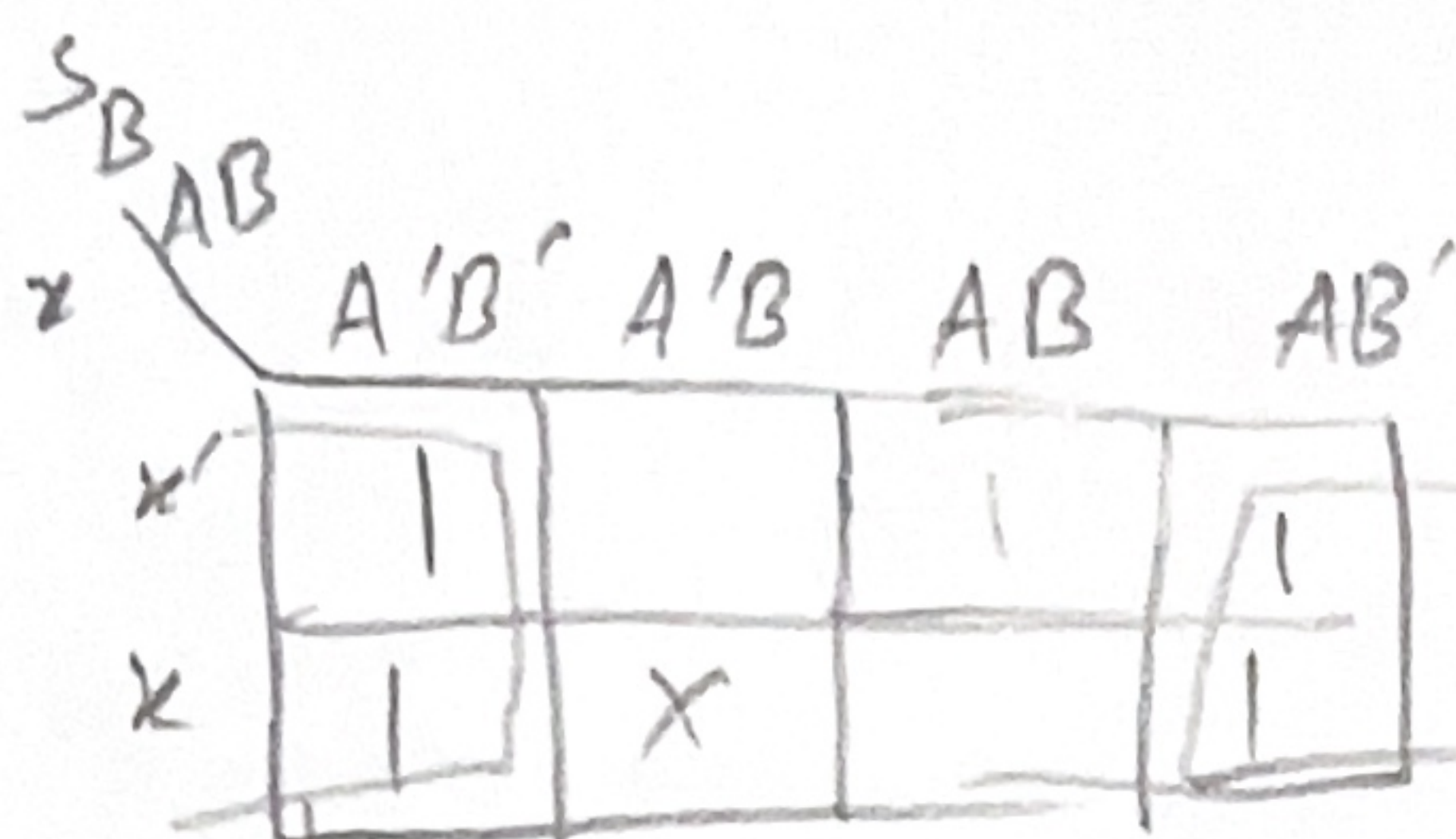
K-Map



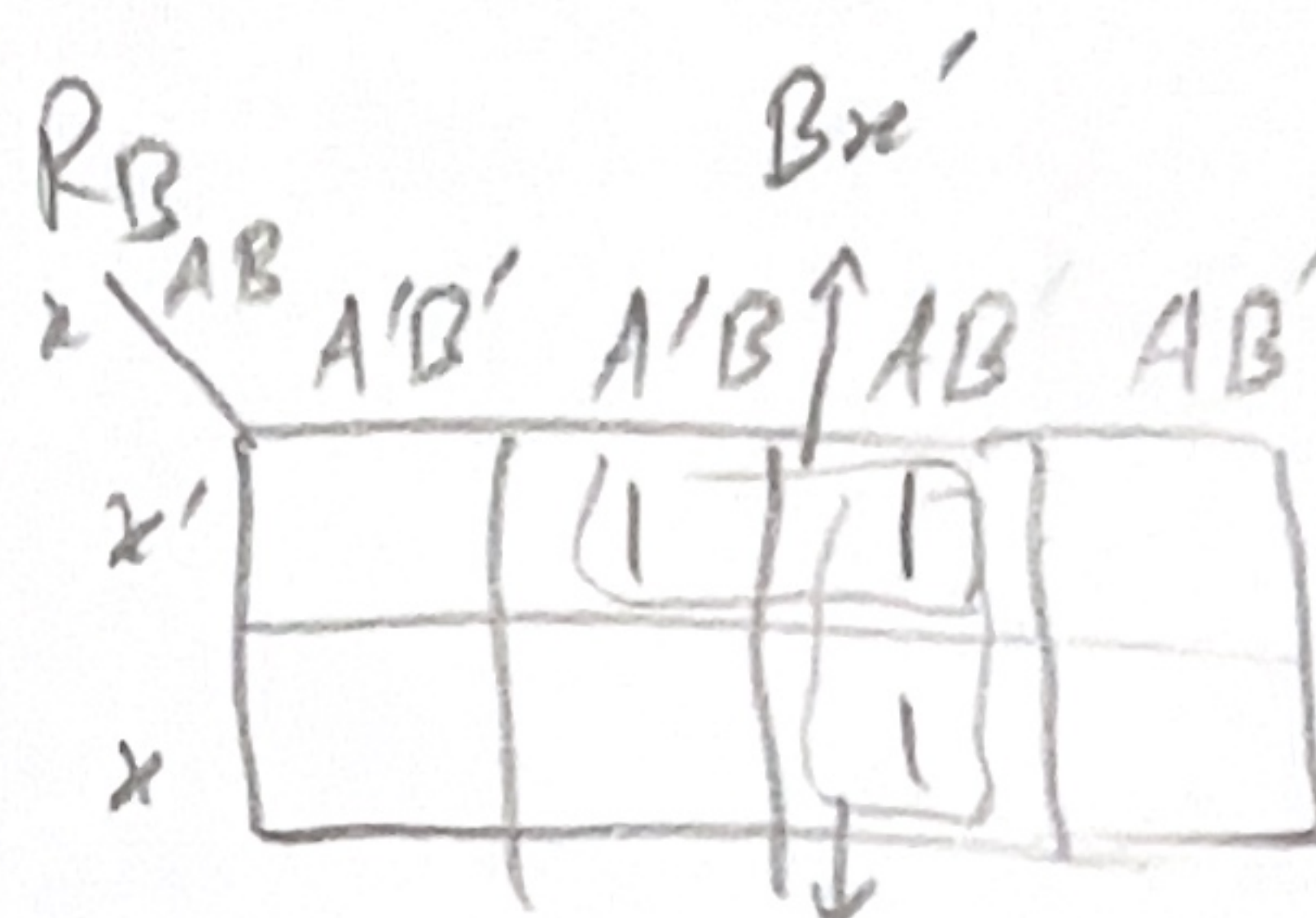
$$S_A = A'B$$



$$R_A = A$$



$$S_B = B'$$



$$R_B = AB + Bz'$$

Circuit diagram:

